

EEE 359

Engineering Project Management

Key Challenges for Organization

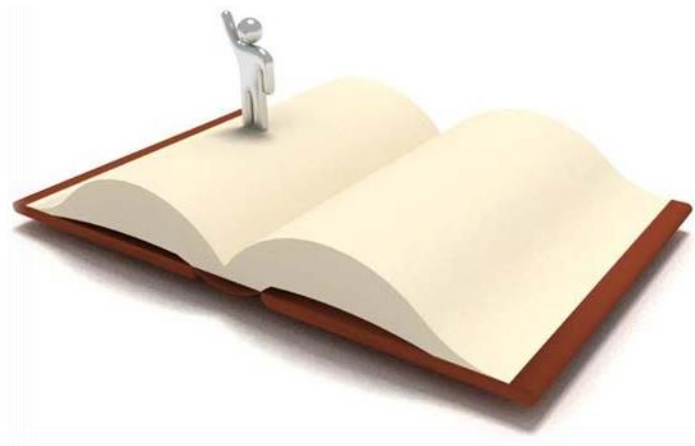
- **Maintain current business operations** (i.e. maintain profitability, service quality, customer relationships, brand loyalty, productivity, market confidence, etc.). This is what we would term 'business as usual'
- **Transform business operations** in order **to survive and compete** in the **future** (i.e. looking forward and deciding how business change can be introduced to best effect for the organization).

As the **pace of change** (technology, business, social, regulatory, etc.) **accelerates**, and the **penalties of failing to adapt to change** become more evident, the focus of management attention is **inevitably moving** to achieve a **balance between** business as usual and business change.

What is Project?

Definition of Project

- A **temporary** endeavor with a **beginning** and an **end**
- Creates a **unique** product, service, or result (P,S,R)



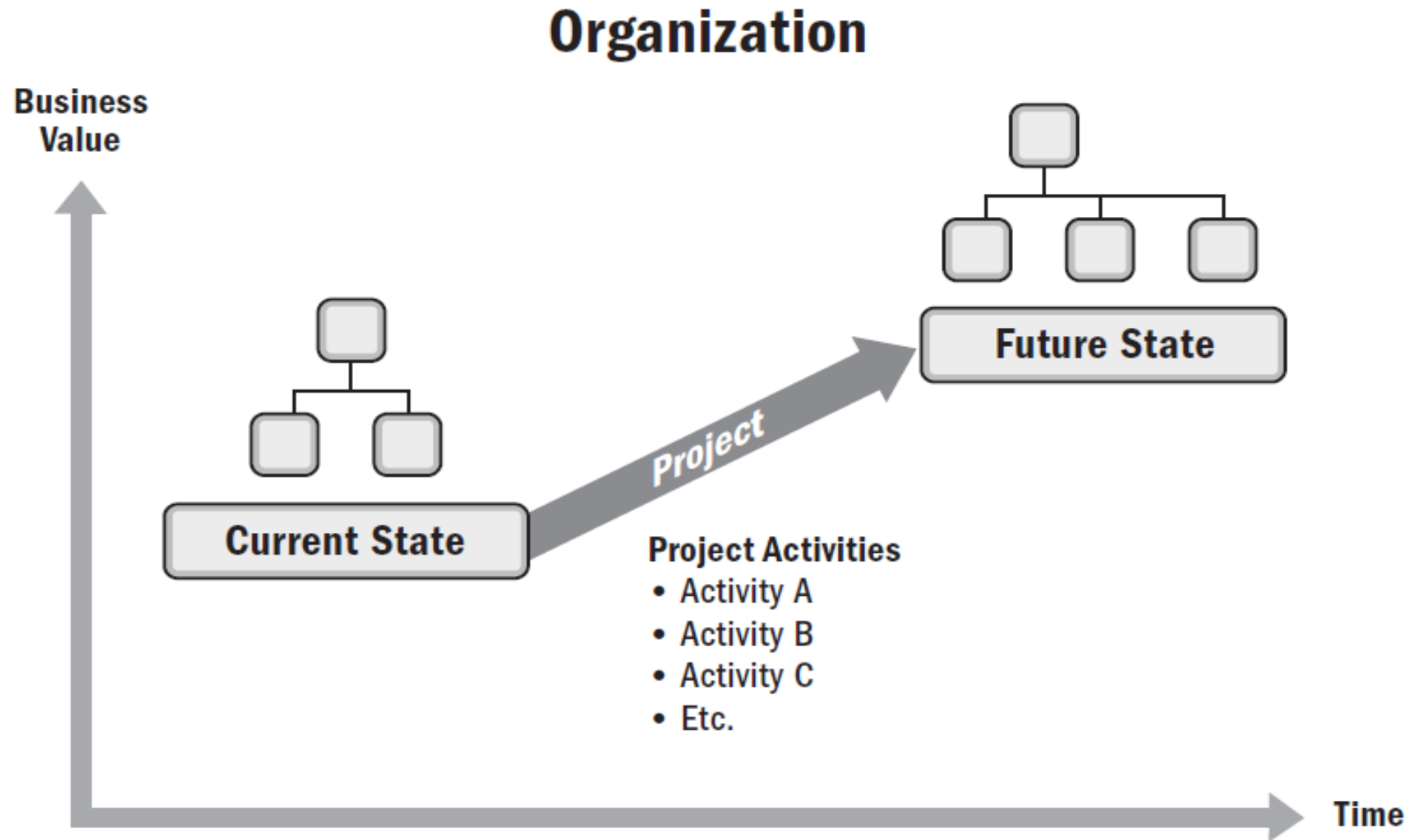
Characteristics of Projects

- **Change:** Projects are the means by which we introduce change
- **Temporary:** As the definition of a project states, projects are temporary in nature. When the desired change has been implemented, business as usual resumes (in its new form) and the need for the project is removed. Projects should have a defined start and a defined end
- **Cross-functional:** A project involves a team of people with **different skills** working together (on a temporary basis) **to introduce a change** that will impact others outside the team. Projects often cross the normal functional divisions within an organization and sometimes span entirely different organizations. This frequently causes stresses and strains both within organizations and between them (for example, between customers and suppliers). Each has a different perspective and motivation for getting involved in the change.

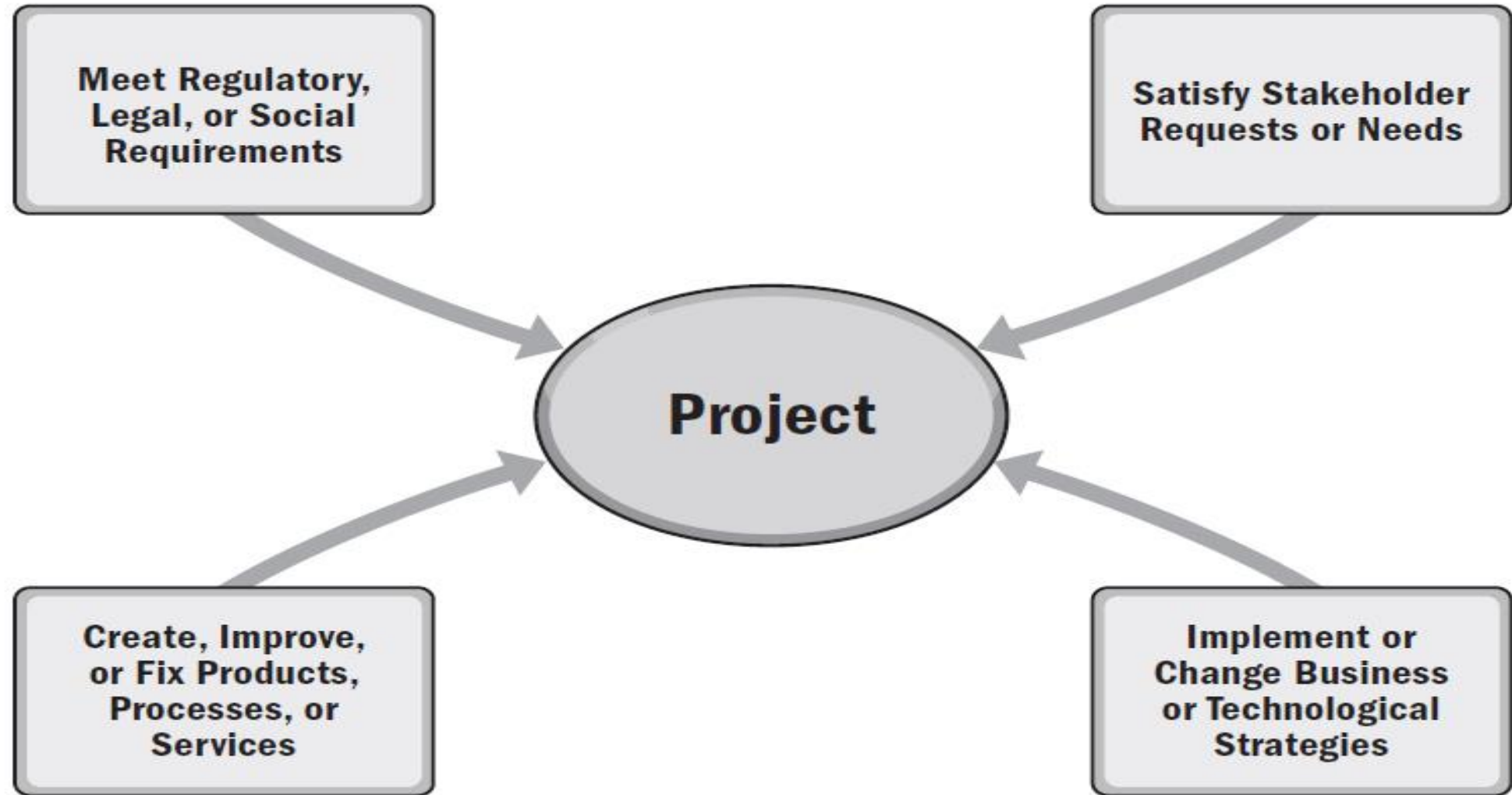
Characteristics of Projects

- **Unique:** Every project is **unique**. An organization may undertake many similar projects, and establish a familiar, proven pattern of project activity, but each one will be unique in some way: a different team, a different customer, a different location, a different time. All these factors combine to make every project unique
- **Uncertainty** The characteristics already listed will introduce **threats** and **opportunities** over and above those we typically encounter in the course of business as usual. **Projects are more risky**

Project Drive Change



Project Initiation Context



Why need Project Management?

Use of Project Management

There is a need for formalized project management when you can answer “**yes**” to any of these questions:

- Is the project technically complex?
- Is there a dynamic or stringent regulatory consideration?
- Are there budgets and/or scheduling constraints?
- Are there several activities to be integrated?
- Are there several functional boundaries to be crossed?

Definition of Project Management

Project management is the **application of knowledge, skills, and tools and techniques** to project activities to **meet project requirements**. Project management refers to guiding the project work to **deliver** the intended outcomes. Project teams can **achieve the outcomes** using a broad range of **approaches** (predictive, hybrid, adaptive).

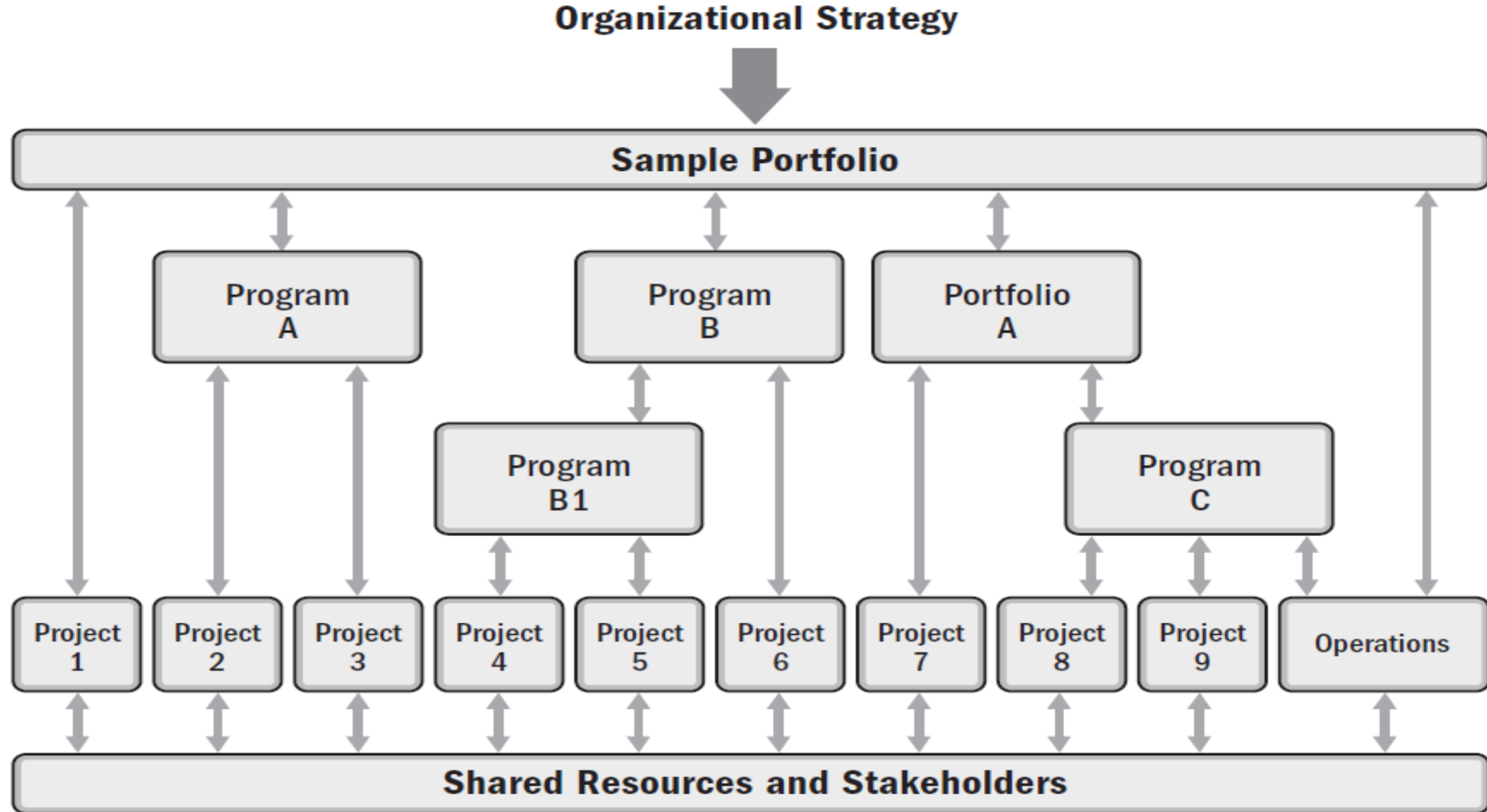
For example, a new house is completed by creating drawings, foundations, floors, walls, windows, a roof, plumbing, wiring and connected services. None of this is project management, so why do we need project management at all? The purpose of project management is to keep control over the specialist work required to create the project's products or, to continue with the house analogy, to make sure the roofing contractor does not arrive before the walls are built.

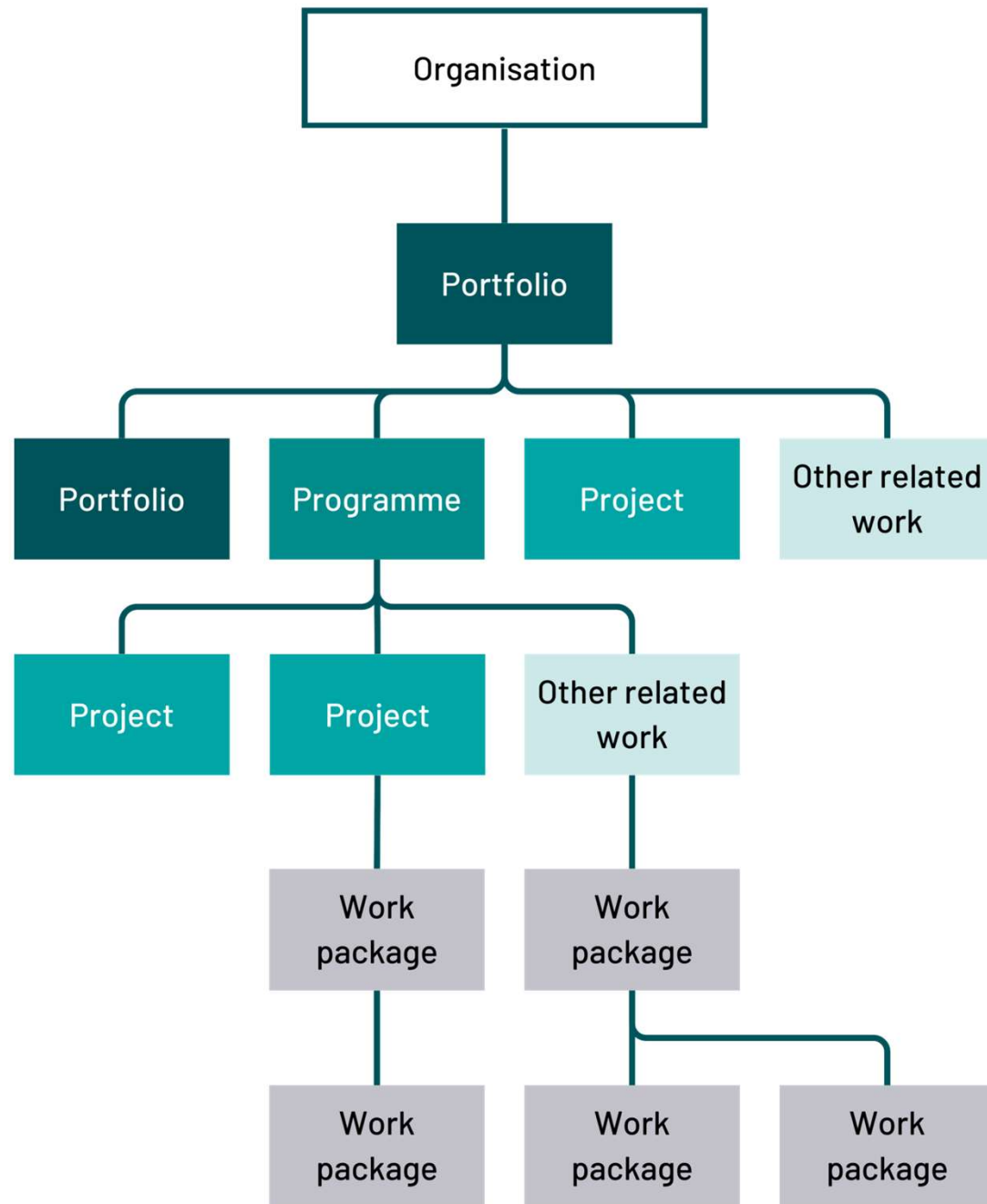
Relationship between Portfolio, Program and Project

Relationship

- Portfolio - refers to a collections of **non-related** programs, projects, subportfolios, and **operations** managed as a group to achieve **strategic objectives**.
- Programs - are grouped within a portfolio, a group of **related** projects, subprograms, and program activities managed in a **coordinated way** to obtain **benefits not available from managing them individually**.
- Projects - can be either within or outside of a program are **still part of a portfolio**.

Relationship





Comparative Overview

	Project	Program	Portfolio
Definition	A project is a temporary endeavor undertaken to create a unique product, service, or result.	A program is a group of related projects, subsidiary programs, and program activities that are managed in a coordinated manner to obtain benefits not available from managing them individually.	A portfolio is a collection of projects, programs, subsidiary portfolios, and operations managed as a group to achieve strategic objectives.
Scope	Projects have defined objectives. Scope is progressively elaborated throughout the project life cycle.	Programs have a scope that encompasses the scopes of its program components. Programs produce benefits to an organization by ensuring that the outputs and outcomes of program components are delivered in a coordinated and complementary manner.	Portfolios have an organizational scope that changes with the strategic objectives of the organization.

Comparative Overview

	Project	Program	Portfolio
Change	Project managers expect change and implement processes to keep change managed and controlled.	Programs are managed in a manner that accepts and adapts to change as necessary to optimize the delivery of benefits as the program's components deliver outcomes and/or outputs.	Portfolio managers continuously monitor changes in the broader internal and external environments.
Planning	Project managers progressively elaborate high-level information into detailed plans throughout the project life cycle.	Programs are managed using high-level plans that track the interdependencies and progress of program components. Program plans are also used to guide planning at the component level.	Portfolio managers create and maintain necessary processes and communication relative to the aggregate portfolio.

Comparative Overview

	Project	Program	Portfolio
Management	Project managers manage the project team to meet the project objectives.	Programs are managed by program managers who ensure that program benefits are delivered as expected, by coordinating the activities of a program's components.	Portfolio managers may manage or coordinate portfolio management staff, or program and project staff that may have reporting responsibilities into the aggregate portfolio.
Monitoring	Project managers monitor and control the work of producing the products, services, or results that the project was undertaken to produce.	Program managers monitor the progress of program components to ensure the overall goals, schedules, budget, and benefits of the program will be met.	Portfolio managers monitor strategic changes and aggregate resource allocation, performance results, and risk of the portfolio.

Comparative Overview

	Project	Program	Portfolio
Success	Success is measured by product and project quality, timeliness, budget compliance, and degree of customer satisfaction.	A program's success is measured by the program's ability to deliver its intended benefits to an organization, and by the program's efficiency and effectiveness in delivering those benefits.	Success is measured in terms of the aggregate investment performance and benefit realization of the portfolio.

EXAMPLES:

Retail

Creation of
a new electric drill.

Creation of a line
of power tools.

All projects and programs
regarding the products
sold by a business.

Construction

Building a house.

All houses built
in a given area.

All projects and programs
of the business.

Software

A new feature
of the software.

An update containing
multiple features
of the software.

All projects and programs
concerning the product.

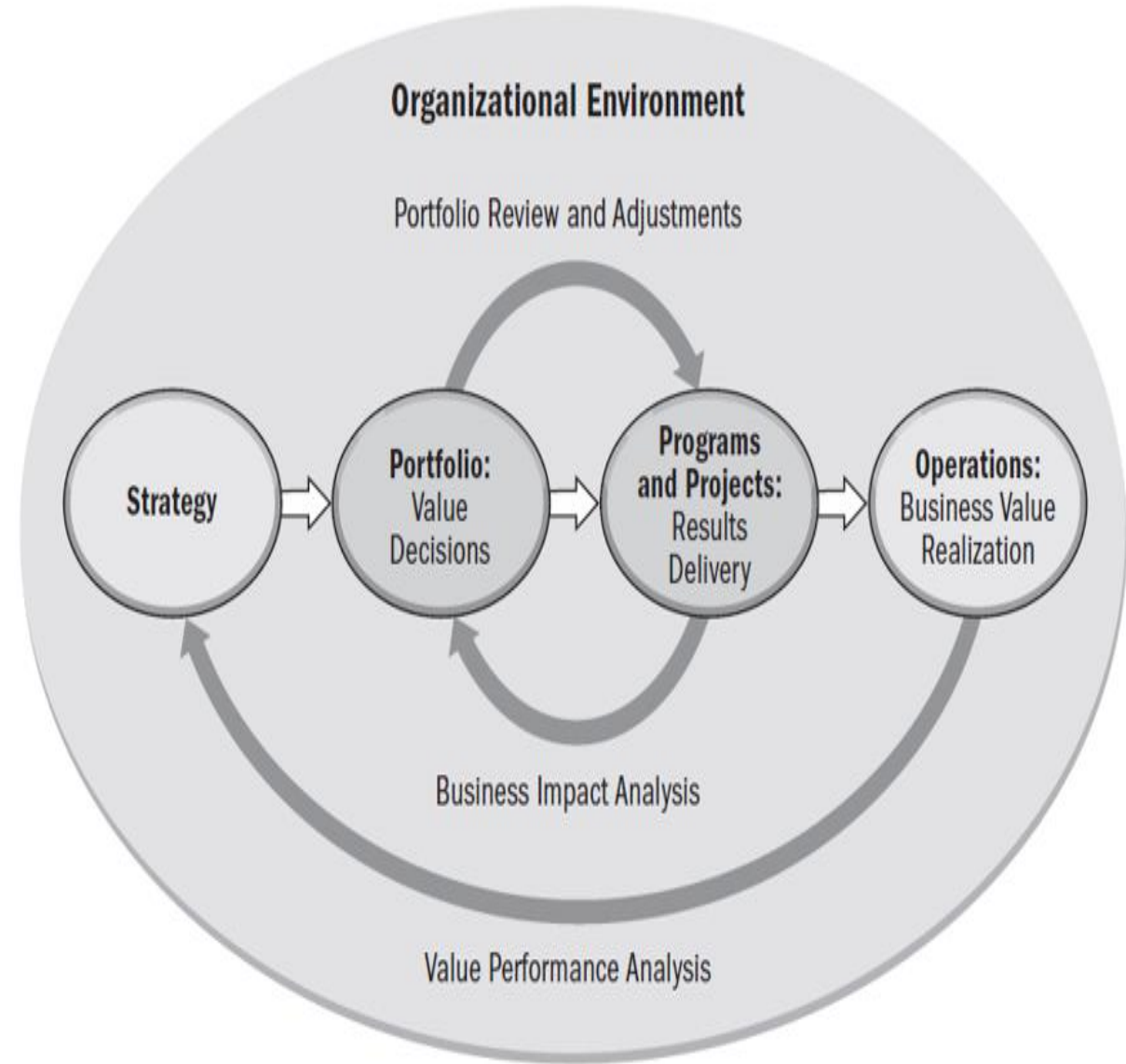
Operation

Operational work is **ongoing** and produces **repetitive** products, services, or results, whereas project work is temporary and ends.



Organizational Project Management (OPM)

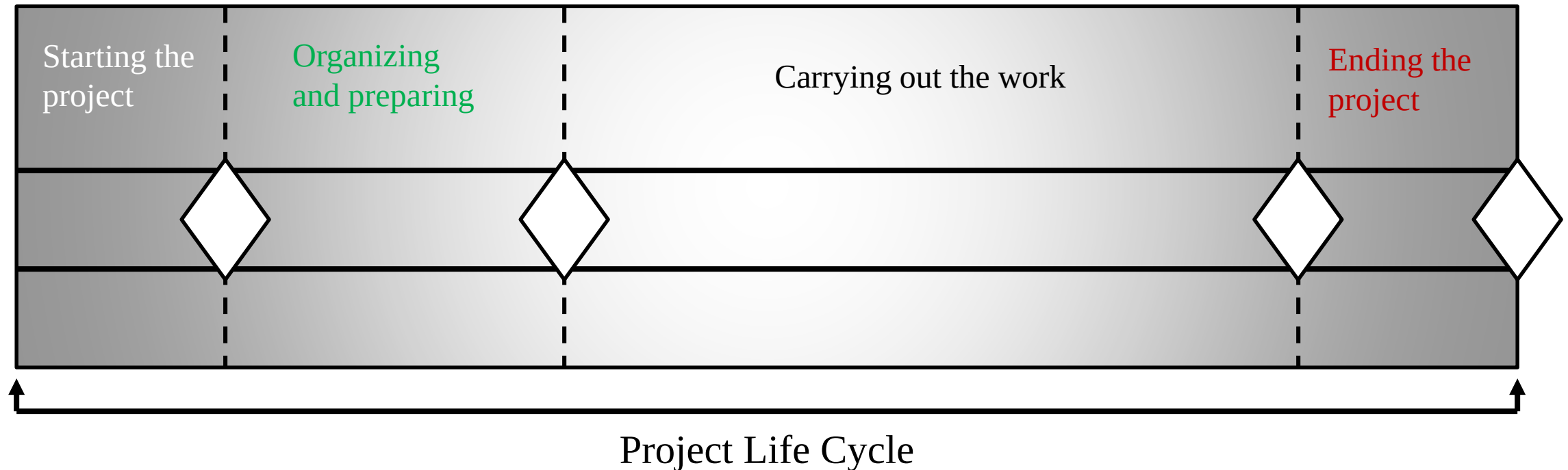
- OPM is defined as a framework in which portfolio, program, and project management are integrated with organizational enablers in order to achieve strategic objectives.
- The purpose of OPM is to ensure that the organization undertakes the right projects and allocates critical resources appropriately.
- OPM also helps to ensure that all levels in the organization understand the strategic vision, the initiatives that support the vision, the objectives, and the deliverables.



Project Life Cycle

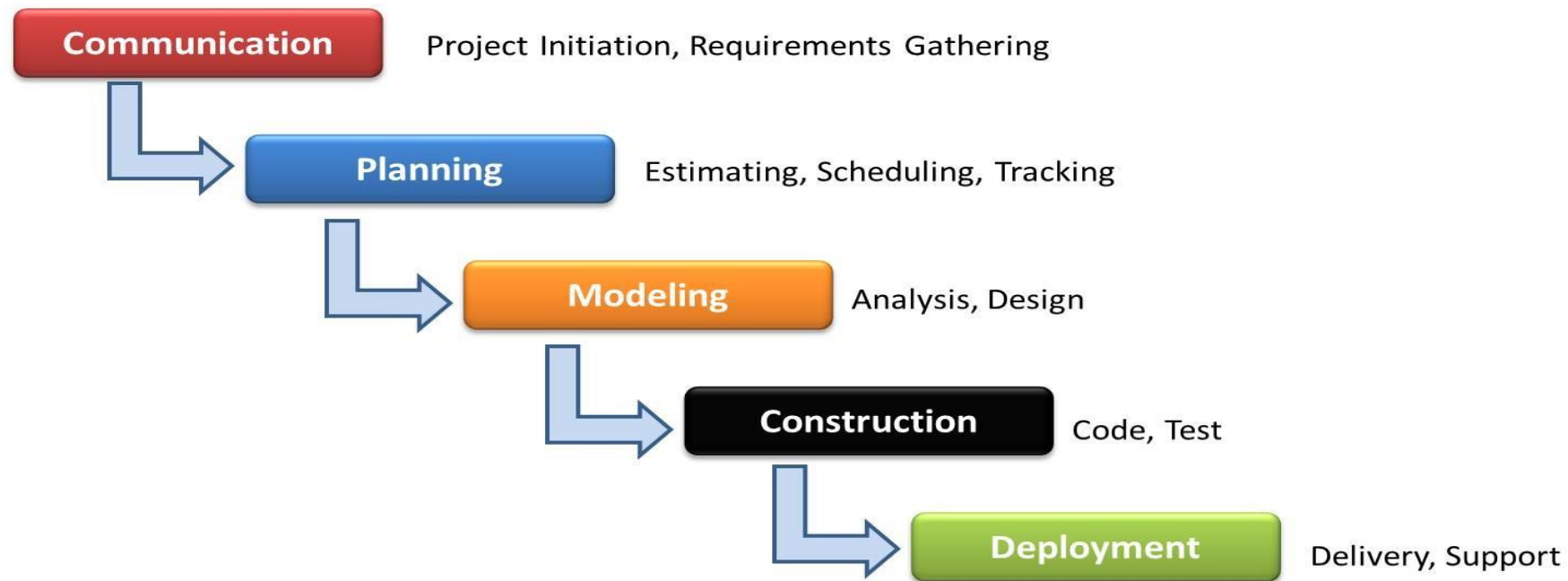
Project Life Cycle

A project life cycle is the **series of phases** that a project passes through from its **start** to its **completion**. It provides the **basic framework** for managing the project. This basic framework applies regardless of the specific project work involved.

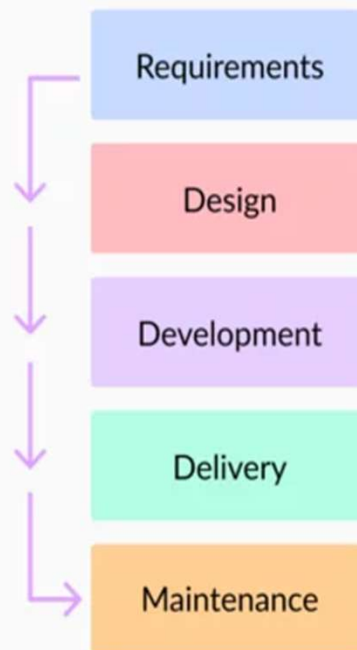


Predictive Life Cycle

In a predictive life cycle, the project scope, time, and cost are **determined in the early phases** of the life cycle. **Any changes** to the scope are **carefully managed**. Predictive life cycles may also be referred to as **waterfall life cycles**.



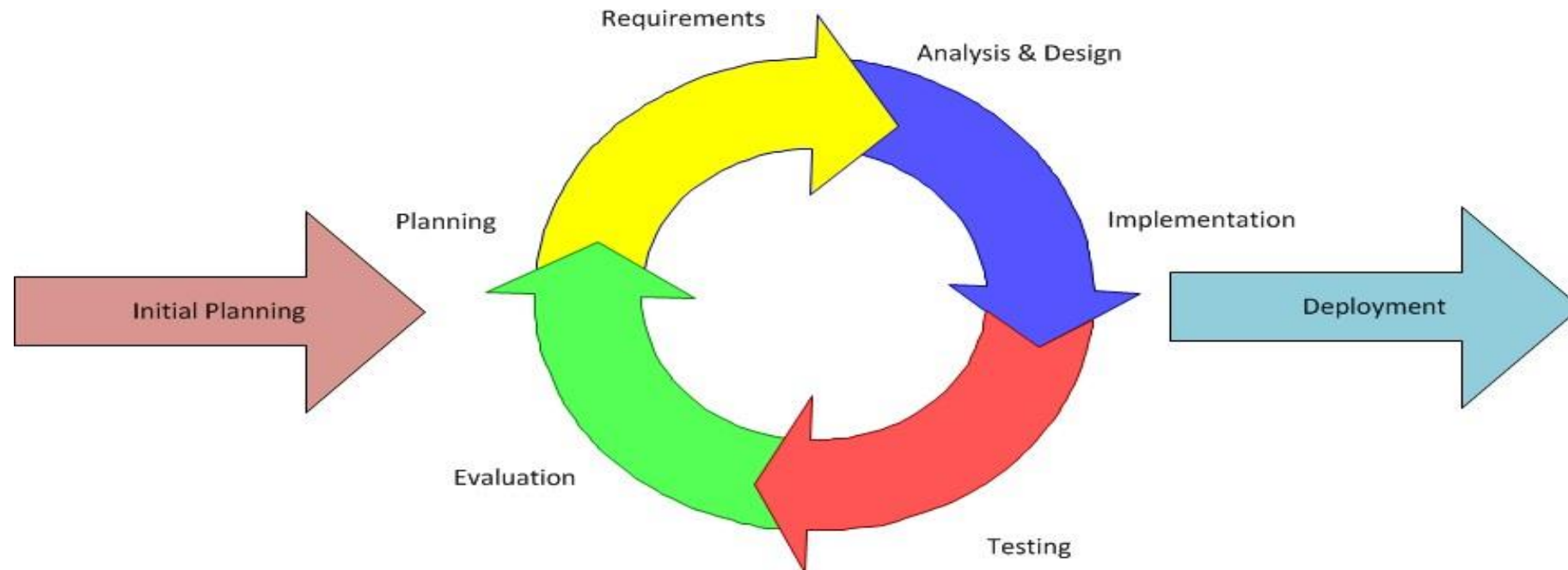
The Waterfall Model



Iterative Life Cycle

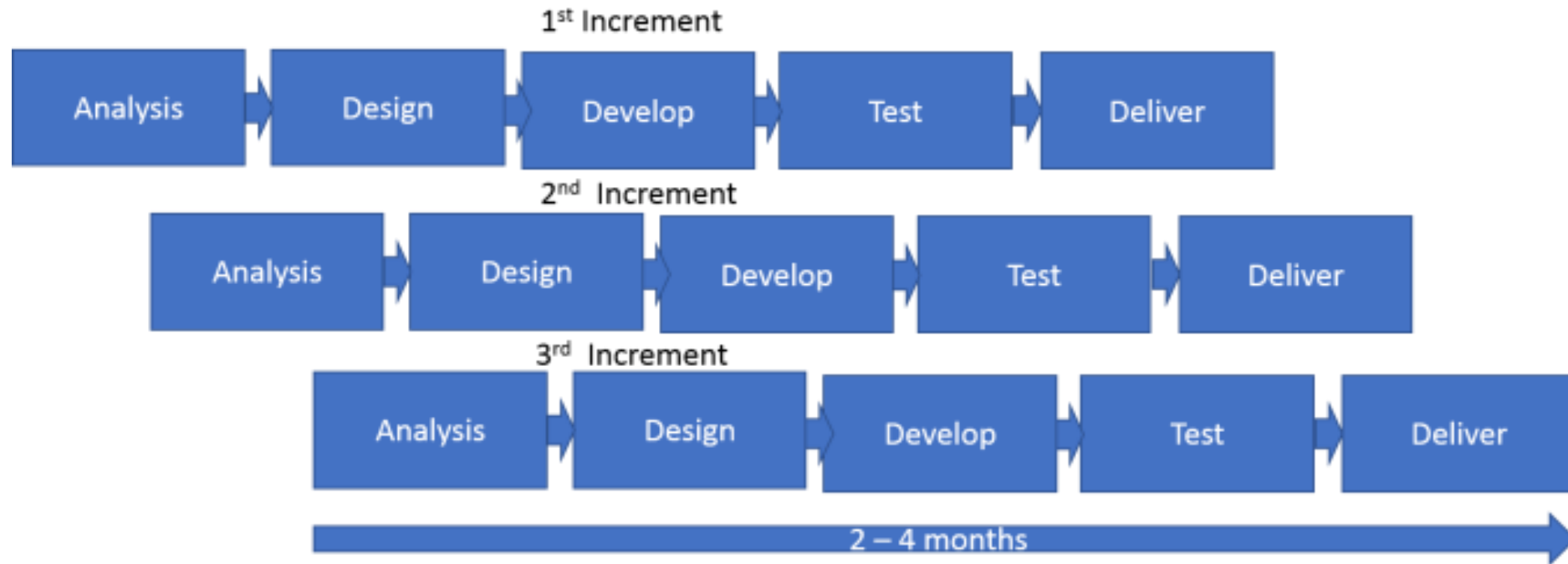
In an iterative life cycle, the project scope is generally **determined early** in the project life cycle, but time and cost estimates are **routinely modified** as the project team's **understanding** of the product **increases**.

Iterations develop the product through a series of repeated cycles, while increments successively add to the functionality of the product.



Incremental Life Cycle

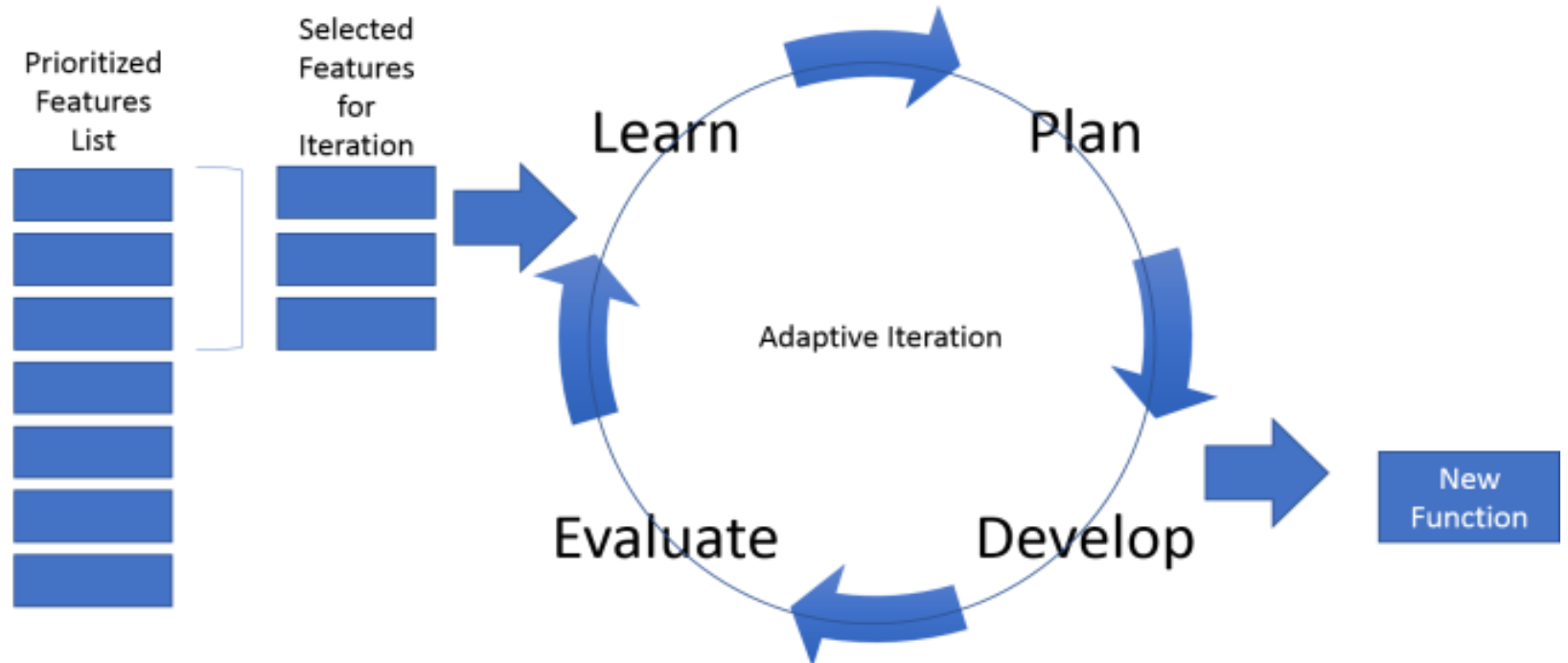
In an incremental life cycle, the deliverable is produced through a **series of iterations** that successively add functionality within a **predetermined time frame**. The deliverable contains the necessary and sufficient **capability** to be considered complete **only after the final iteration**.



Adaptive/Agile Life Cycle

Adaptive life cycles are agile, iterative, or incremental. The detailed scope is defined and approved before the start of an iteration. Adaptive life cycles are also referred to as agile or change-driven life cycles.

- Preferred for rapidly changing environment
- Known as change driven or agile methods
- Rapid iterations, fixed in time and costs
- Uses a product backlog
- Standing Meetings



Hybrid Life Cycle

A hybrid life cycle is a **combination of a predictive and an adaptive life cycle**. Those elements of the project that are well known or have fixed requirements follow a predictive development life cycle, and those elements that are still evolving follow an adaptive development life cycle.

Summary of Life Cycle

Characteristics				
Approach	Requirements	Activities	Delivery	Goal
Predictive	Fixed	Performed once for the entire project	Single delivery	Manage Cost
Iterative	Dynamic	Repeated until correct	Single delivery	Correctness of solution
Incremental	Dynamic	Performed once for a given increment	Frequent small deliveries	Speed
Agile	Dynamic	Repeated until correct	Frequent small deliveries	Customer value via frequent deliveries and feedback

Project Phase

Project Phase

- A Project Phase is a collection of **logically related activities** that culminates in the completion of one or more **deliverables**.
- Phase structure allows the project to be segmented into logical subsets for ease of management, planning, and control.
- The **closure of a phase** has a transfer or handoff of the phase deliverables. May be referred to as a **phase gate, phase review or kill point** to allow change or termination.

Project Phase

The project phases may be established based on various factors:

- Management needs
- Nature of the project
- Unique characteristics of the organization, industry, or technology
- Project elements including, but not limited to, technology, engineering, business
- Decision points (e.g., funding, project go/no-go, and milestone review)

The 5 phases of the project life cycle

01

Initiation

Initiation is the project kick-off phase. This is when you collect all the information needed from key stakeholders. This phase defines the problem and details what needs to be done.

02

Planning

The planning phase is the foundation for the project. This is when you detail all the work, how you plan on doing it, and what resources you'll need to meet project goals.

03

Execution

The execution phase is when you get stuff done. This is where you put everything you planned into motion and complete deliverables to achieve the project objectives.

04

Monitoring

The monitoring phase goes hand-in-hand with the execution phase. It's about monitoring the project's overall progress to see if you're meeting project objectives and are on track.

05

Closure

The closure phase wraps things up. You'll hand over documentation, release final deliverables, terminate contracts, and inform everyone that the project is complete.

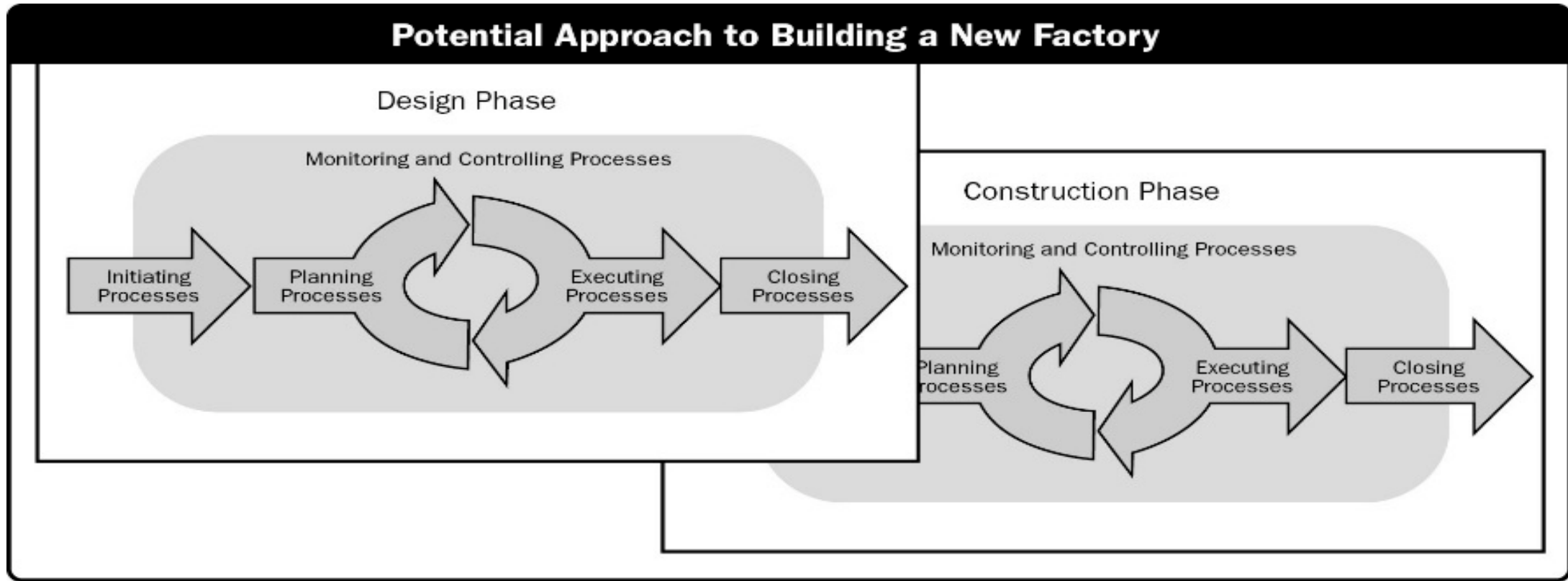
Phase to Phase Relationship

Sequential Phase: A phase starts only when the previous phase is complete.



Phase to Phase Relationship

Overlapping Phase: A phase starts prior to the completion of the previous one.



Thanks